Eduard Dehelean

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Education

Minerva University — San Francisco, CA
Bachelor of Science, Computer Science — *2026 – Present*

Calvin University — Grand Rapids, MI
Bachelor of Computer Science, Minor in Data Science — *Aug. 2024 – May 2026*

Liviu Rebreanu National College — Bistrita, Romania
Math & Computer Science — *Sep 2020 – May 2024*

Experience

Undergraduate Research Assistant — June 2025 – Present
Calvin University (under CAIDA, UC San Diego) — *Remote (Romania) / On-site (Grand Rapids, MI)*

- Conducted a longitudinal study of Geofeed adoption, data quality, and dynamics using a large-scale dataset collected daily over 13+ months from all five Regional Internet Registries
- Developed Python-based data processing and validation pipelines to extract, normalize, and verify Geofeed links and entries in compliance with RFC 8805 and RFC 9092
- Performed link-level and entry-level deduplication across IPv4 and IPv6 datasets, identifying systemic duplication patterns and their impact on adoption metrics
- Analyzed Geofeed entries by ASNs, inetnum/inet6num objects, and upstream ISPs to identify structural sources of data growth and inconsistency

Computer Science Grader & AI Tool Developer — Sep 2025 – Present
Calvin University — *Grand Rapids, MI*

- Graded assignments for computing courses, ensuring accurate evaluation and high-quality feedback at scale
- Built a React and GPT-based tool to automate manual Moodle grading, batching submissions and generating structured feedback to cut grading time and improve consistency

Computer Science Tutor — Sep. 2025 – Present
Calvin University — *Grand Rapids, MI*

- Tutored students in introductory Computer Science and Data Structures & Algorithms courses, supporting both high-achieving students and those recovering from academic difficulties
- Explained complex programming and algorithmic concepts in a clear and intuitive manner, adapting explanations to individual learning styles
- Demonstrated advanced mastery of CS fundamentals and effective pedagogical skills by guiding students to confidently pass and excel in coursework

Projects

ClearPath — Accessibility Driving Assistant | *Python, FastAPI, YOLOv8, OpenCV, GPT-4o Vision*

- Built the backend and CV pipeline for ClearPath, winner of the Called to Create hackathon — a real-time assistant that reads traffic lights, road signs, and hazards aloud for colorblind and aging drivers via phone camera or dashcam
- Developed a FastAPI WebSocket service running YOLOv8 on live video frames, with HSV color analysis in OpenCV to classify red, yellow, and green signals more reliably than object detection alone
- Added road-sign reading via a secondary detection pass to GPT-4o Vision and pedestrian/cyclist alerts on green-light transitions, delivered through redundant audio and high-contrast, colorblind-safe visual cues

AI Grading Feedback Tool | *React, JavaScript, GPT API, Moodle*

- Developed a React-based AI-assisted tool to streamline grading workflows for introductory computing courses
- Implemented batch processing of assignments and integrated a GPT-based API to generate structured, code-aware feedback
- Automated CSV generation with per-student scores and comments for instructor review and LMS integration (work in progress)

Pixel Craft — Python Image Processing Application | *Python, PySide6, Pillow, NumPy*

- Developed a Python desktop application for pixel-art generation featuring grid-based color averaging, a real-time PySide6 GUI for pixel size control, NumPy-optimized RGB computation, image transformations, and local export functionality